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Robotic endoscopy system

Abstract

A robotic medical device designed for tissue extraction, featuring a modular construction and motion control assembly. The system includes a robotic arm with multiple joints providing at least three degrees of freedom, a main body with a motorized vacuum pump, and a user interface for operational control. A needle assembly, secured within a removable proximal hub, incorporates a hollow needle with one or more cutting features, a geared wheel for transmitting oscillatory and translational motion, and a luer hub with an O-ring to maintain vacuum pressure. A suction tube connects the needle assembly to the vacuum pump, maintaining negative pressure. The device includes features such as a disposable and reusable elements, quick-connect mechanisms, and integrated sensors for monitoring motion and pressure.

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Background/Summary

BACKGROUND OF THE INVENTION

1. Field of the Invention
- (1) This invention relates, generally, to medical devices. More specifically, it relates to a robotic platform for controlling a biopsy syringe.
2. Brief Description of the Prior Art
- (2) The accurate diagnosis of various medical conditions, including cancer, often requires obtaining tissue samples through biopsy procedures. Biopsies involve the extraction of tissue from a suspected site within the body for histopathological analysis. Current approaches to performing biopsies typically fall into one of two categories: manual and semi-automated. While these methods have seen significant advancements, they still present notable challenges that impact procedural accuracy, efficiency, and patient outcomes.
- (3) Manual biopsy techniques rely heavily on the clinician's skill and experience. This approach involves visual or imaging guidance, such as ultrasound or CT scans, to position a biopsy needle at the target site. In addition, the clinician needs to manipulate the needle in a specific manner to capture tissue from the target site. Despite the widespread use of these methods, there are inherent limitations. Precise targeting and acquisition can be difficult, particularly for deep-seated or small lesions, due to natural physiological movement (e.g., breathing or heartbeat) or limited visibility in imaging modalities. This lack of precision can lead to inadequate or inaccurate sampling, requiring repeat procedures and causing additional discomfort to patients.
- (4) Semi-automated biopsy systems incorporate devices designed to assist in needle insertion and sample collection. These devices often include mechanisms to improve needle stabilization or provide mechanical assistance during the biopsy. However, these systems still rely heavily on manual operation for needle placement and alignment. The lack of real-time adaptive feedback and precision limits their utility in complex procedures, especially in cases requiring millimeter-level accuracy.
- (5) The emergence of robotic-assisted medical technologies has highlighted the potential to address

the deficiencies of traditional biopsy techniques. Robotic systems offer unparalleled control, precision, and stability in minimally invasive procedures. Despite these advantages, the integration of robotic systems into biopsy procedures remains underexplored due to inherent challenges associated with integration of biopsy technology and the considerations of multi-use components, replacement components, and ease of use.

(6) Accordingly, what is needed is a robot-controlled biopsy needle system that combines multi-directional movement control with advanced imaging and feedback mechanisms. This innovation represents a significant step forward in the field of medical diagnostics, offering a solution to the challenges of precision, consistency, and procedural safety that have long plagued biopsy practices. However, in view of the art considered as a whole at the time the present invention was made, it was not obvious to those of ordinary skill in the field of this invention how the shortcomings of the prior art could be overcome.

SUMMARY OF THE INVENTION

(7) The long-standing but heretofore unfulfilled need for an advanced robotic biopsy device that combines oscillatory and translational movement is now met by a novel and non-obvious invention disclosed and claimed herein.

(8) The present invention includes a robotic medical device comprising a needle assembly that includes a hollow needle, a needle gear at the proximal end of the hollow needle, and a luer hub with an O-ring, wherein the needle is configured to rotate relative to the luer hub. The invention further includes a robotic arm including multiple joints providing at least three degrees of freedom, a translation guide and a needle translator configured to axially translate the needle assembly along the length of the translation guide, a motor operably engaging the needle gear to rotate the needle about its longitudinal axis, and a tissue chamber established at least in part by a hollow interior of the needle and in fluidic communication with a vacuum source via a suction tube. The needle is configured to be axially translated and rotated relative to its longitudinal axis while the tissue chamber is subject to the vacuum.

(9) In an embodiment, the invention further comprises a proximal hub configured to house the needle assembly and temporarily attach to the needle translator. The proximal hub comprises a removable attachment mechanism, a luer hub retention latch, an interface for transferring oscillatory and translational motion from the motor to the geared wheel, and an anti-buckling tube.

(10) In an embodiment, the invention further comprises a distal hub configured to be temporarily coupled to the terminal joint of the robotic arm. The distal hub comprises a removable attachment mechanism, a sheath clamping latch, a guide tube, and an endoscope.

(11) In an embodiment, the needle assembly and suction tube are disposable. In an embodiment, the robotic arm includes a detachable joint mechanism with alignment pins and a quick-release locking collar to facilitate removal and reattachment.

(12) In an embodiment, the invention further includes sensors, including a position sensor configured to monitor the movement of the robotic arm, a pressure sensor configured to monitor vacuum levels within the tissue chamber, and a force-torque feedback sensor to monitor the rotation and translation of the needle.

(13) In an embodiment, the invention includes a robotic medical device comprising a needle assembly that includes a hollow needle and a needle gear disposed near the proximal end of the needle. The invention further includes a robotic arm, the robotic arm including a translation guide and a needle translator, a proximal hub configured to house the needle and temporarily attach to the needle translator, a motor configured to operably engage the needle gear to rotate the needle about its longitudinal axis, and a distal hub configured to be temporarily coupled to the terminal joint of the robotic arm. The needle translator is configured to translate the needle through the distal hub and translate the proximal hub towards the distal hub. The distal hub includes a passage for receiving the needle, the needle including a tissue chamber established at least in part by the hollow interior of the needle and in fluidic communication with a vacuum source via a suction tube. The

needle is configured to be axially translated and rotated relative to its longitudinal axis while the tissue chamber is subject to a vacuum.

(14) In an embodiment, the proximal hub comprises a removable attachment mechanism, a luer hub retention latch, and an interface for transferring oscillatory and translational motion from the motor to the geared wheel.

(15) In an embodiment, the invention further includes an anti-buckling tube extending at least partially between the distal hub and the proximal hub, wherein the anti-buckling tube has greater rigidity than the needle.

(16) In an embodiment, the distal hub comprises a removable attachment mechanism, a sheath clamping latch, a guide tube, and an endoscope.

(17) In an embodiment, the needle assembly and suction tube are disposable.

(18) In an embodiment, the robotic arm includes a detachable joint mechanism with alignment pins and a quick-release locking collar to facilitate removal and reattachment.

(19) In an embodiment, the device further comprises sensors, including a position sensor configured to monitor movement of the robotic arm, a pressure sensor configured to monitor vacuum levels within the tissue chamber, and a force-torque feedback sensor configured to monitor the rotation and translation of the needle.

(20) In an embodiment, the invention provides a robotic medical device comprising a needle assembly that includes a hollow needle, a needle gear disposed proximate to the proximal end of the needle, a robotic arm including a translation guide and a needle translator, a proximal hub configured to house the needle and temporarily attach to the needle translator, a motor operably engaging the needle gear to rotate the needle about its longitudinal axis, and a distal hub temporarily coupled to the terminal joint of the robotic arm, wherein the distal hub includes a passage for receiving the needle. The needle translator is configured to translate the needle through the distal hub and translate the proximal hub towards the distal hub. The tissue chamber is established at least in part by the hollow interior of the needle and is in fluidic communication with a vacuum source. The needle is configured to be axially translated and rotated relative to its longitudinal axis while the tissue chamber is subject to a vacuum.

(21) In an embodiment, the proximal hub comprises a removable attachment mechanism, a luer hub retention latch, and an interface for transferring oscillatory and translational motion from the motor to the geared wheel.

(22) In an embodiment, the device further includes an anti-buckling tube extending at least partially between the distal hub and the proximal hub, wherein the anti-buckling tube has greater rigidity than the needle.

(23) In an embodiment, the distal hub comprises a removable attachment mechanism, a sheath clamping latch, a guide tube, and an endoscope.

(24) In an embodiment, the device further comprises sensors, including a position sensor configured to monitor movement of the robotic arm, a pressure sensor configured to monitor vacuum levels within the tissue chamber, and a force-torque feedback sensor configured to monitor the rotation and translation of the needle.

(25) Additional aspects and advantages of the present disclosure will become readily apparent to those skilled in this art from the following detailed description, wherein only illustrative embodiments of the present disclosure are shown and described. As will be realized, the present disclosure is capable of other and different embodiments, and its several details are capable of modifications in various obvious respects, all without departing from the disclosure. Accordingly, the drawings and description are to be regarded as illustrative in nature, and not restrictive.

(26) The invention accordingly comprises the features of construction, combination of elements, and arrangement of parts that will be exemplified in the disclosure set forth hereinafter and the scope of the invention will be indicated in the claims.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) For a fuller understanding of the invention, reference should be made to the following detailed description, taken in connection with the accompanying drawings, in which:
- (2) FIG. 1 is a perspective view of an embodiment of the present invention;
- (3) FIG. 2 is a close-up perspective view of an embodiment of the robotic arm;
- (4) FIG. 3 is a focused view of the distal portion of the robot arm, including the terminal joint and distal housing unit;
- (5) FIG. 4 is a block diagram of an embodiment of a first motor control circuit in which oscillation of the motor is effected by timing;
- (6) FIG. 5 is a block diagram of a first motor control circuit in which oscillation of the motor is effected by a motor position sensor;
- (7) FIG. 6 is a flow chart of logic implemented by the circuit of FIG. 4;
- (8) FIG. 7 is a flow chart of logic implemented by the circuit of FIG. 5;
- (9) FIG. 8 is a perspective view of the needle assembly;
- (10) FIG. 9 is an x-ray view of the proximal hub;
- (11) FIG. 10 is an x-ray view of the distal hub;
- (12) FIG. 11 is an exploded view of the needle assembly, proximal hub, and distal hub;
- (13) FIG. 12 is a perspective view robot arm as assembled prior to operation;
- (14) FIG. 13 is an x-ray view of the proximal hub coupled to the housing unit with the needle assembly and suction tubing in place;
- (15) FIG. 14 is an x-ray view of the robot arm as assembled prior to operation;
- (16) FIG. 15 depicts an embodiment of a gear assembly for an axial oscillation mechanism for the needle;
- (17) FIG. 16 depicts a partial exploded view of an embodiment of an axial oscillation mechanism and the needle.
- (18) FIG. 17 is a diagram illustrating the axial oscillation of the needle as the gear assembly rotates.
- (19) FIG. 18 depicts an embodiment of a gear assembly for an axial oscillation mechanism for the needle;
- (20) FIG. 19 depicts an embodiment of a gear assembly for an axial oscillation mechanism for the needle;
- (21) FIG. 20 depicts an embodiment of a gear assembly for an axial oscillation mechanism for the needle; and
- (22) FIG. 21 illustrates an axial oscillation mechanism for the needle.

DETAILED DESCRIPTION OF THE INVENTION

(23) In the following detailed description of the preferred embodiments, reference is made to the accompanying drawings, which form a part thereof, and within which are shown by way of illustration specific embodiments by which the invention may be practiced. It is to be understood that one skilled in the art will recognize that other embodiments may be utilized, and it will be apparent to one skilled in the art that structural changes may be made without departing from the scope of the invention.

(24) As such, elements/components shown in diagrams are illustrative of exemplary embodiments of the disclosure and are meant to avoid obscuring the disclosure. Any headings, used herein, are for organizational purposes only and shall not be used to limit the scope of the description or the claims.

(25) Reference in the specification to “one embodiment,” “preferred embodiment,” “an embodiment,” or “embodiments” means that a particular feature, structure, characteristic, or function described in connection with the embodiment is included in at least one embodiment of

the disclosure and may be in more than one embodiment. The appearances of the phrases “in one embodiment,” “in an embodiment,” “in embodiments,” “in alternative embodiments,” “in an alternative embodiment,” or “in some embodiments” in various places in the specification are not necessarily all referring to the same embodiment or embodiments. The terms “include,” “including,” “comprise,” and “comprising” shall be understood to be open terms and any lists that follow are examples and not meant to be limited to the listed items.

(26) The present invention includes a robotic medical device designed for tissue extraction, featuring a modular construction and motion control assembly. The system includes a robotic arm with multiple joints providing at least three degrees of freedom, a main body with a motorized vacuum pump, and a user interface for operational control. The system further includes a needle assembly configured to rotate and translate the needle while maintaining a vacuum through the needle. The device includes features such as a disposable and reusable elements, quick-connect mechanisms, and integrated sensors for monitoring motion and pressure.

(27) Referring now to FIG. 1, the present invention includes a robotic medical device **101** for manipulation of a needle **102** used in medical procedures. The needle **102** may be a hollow metal hypodermic needle of a size of between 18 to 27 gauge, and more particularly 22-25 gauge. The tip of the needle **102** is sharpened to facilitate cutting tissue when the needle **102** is advanced into tissue and rotated. The needle may also include cutting features disposed in the lateral side walls of the needle for cutting and extracting tissue.

(28) The device **101** includes a main body **103** that serves as the central structural and functional unit of the device **101**. The main body **103** is configured to securely house and protect the system's internal components while providing a stable base for interfacing with modular components. The device **101** is mounted on wheels **104** or other appropriate method of locomotion that allows the device to be moved and secured at a desired location.

(29) The main body **103** also houses a power management system that includes a power supply unit that converts input power—whether from an AC source or a battery—into the required voltages for the platform's various subsystems. Voltage regulators may be integrated to maintain consistent output levels, protecting sensitive components from power surges or fluctuations.

(30) Additionally, the main body **103** incorporates ventilation channels and thermal management features, including heat sinks and strategically placed fans, to maintain optimal operating temperatures for sensitive electronic components. The main body **103** also includes integrated cable management systems, allowing for secure and organized routing of electrical connections to minimize interference and improve system reliability.

(31) In some embodiments, the present invention includes a user interface **105** securely connected to the main body **103** that serves as the primary input and output device, allowing the user to control the platform's functions and monitor system status in real time. The user interface **105** may include but is not limited to a touch screen, physical controls mounted on the main body, a handheld controller, and remote operating interface. However, it is contemplated that alternative user interface control devices can be used.

(32) The present invention further includes a vacuum system. The vacuum pump is configured to generate negative pressure for tissue extraction. The vacuum system may be housed within the main body **103** and include suction tubing **109** in fluidic communication with the needle **102** to aid in the extraction of tissue. In an embodiment, the vacuum system may operate under programmable control to modulate suction levels based on procedural requirements. In an embodiment, integrated pressure sensors monitor the system in real time, providing feedback to maintain suction levels and alerting the user to any anomalies.

(33) As best depicted in FIG. 2, the present invention further includes a multi-segmented, articulated robotic arm assembly **106** that is mechanically coupled to the main body **103** at an appropriate position to allow complete freedom of movement without interfering with other components. The needle **102** is mechanically coupled to the distal end of the arm assembly **106**

through a removable proximal hub **107** and removable distal hub **108**. The arm assembly **106** is designed with multiple joints that may provide rotational, vertical, and horizontal motion. In an embodiment, the articulations are configured to allow complete rotational freedom.

(34) In some embodiments, the robotic arm assembly **106** rotatably attached to the main body **103** at a base joint **110**. In one embodiment, the base joint **110** can be configured for detachment from the main body through a modular quick-release system. The arm assembly **106** consists of a series of rigid segments, each connected by joints allowing for movement along at least one of the x-, y-, and z-axis. The joints may include adjustable locking or damping systems to stabilize the arm in its desired position when movement is complete. The base joint **110** connects the proximal end of the arm assembly **106** to the main body **103** and may provide rotational freedom of the arm assembly **106** around the vertical axis.

(35) A shoulder joint **112** is positioned at the distal end of a first segment **114** of the arm assembly **106**, providing a second point of rotation about the vertical axis of the shoulder joint **112**. The configuration allows a second segment **116** of the robot arm assembly **106** to move along its horizontal plane, parallel to the first segment **114**. In some embodiments, the shoulder joint **112** is configured to rotate about its horizontal axis allowing the second segment **116** to move along its vertical plane.

(36) The shoulder joint **112** can be configured as a permanent joint including but not limited to a slewing ring bearing, turntable bearing, rotary coupling, or any other device or mechanism allowing full rotation of the joint. However, in some embodiments, the shoulder joint **112** is configured with a detachable mechanism, allowing the arm to be removed for cleaning and maintenance.

(37) The arm assembly **106** may further include an elbow joint **118** coupled to the distal end of the second segment **116** of the robot arm assembly **106**. The elbow joint **118** connects to the proximal end of a third segment **120** of the robot arm assembly **106**, allowing rotation along the vertical axis of the third segment **120**. In another embodiment, the elbow joint **118** is configured to rotate about its horizontal axis allowing the third segment **120** to move along its vertical plane. The elbow joint **118** can be configured as a permanent joint or as a detachable mechanism.

(38) The distal end of the third segment **120** is connected to a wrist joint **122**. The wrist joint **122** is connected to the proximal end of a fourth segment **124** of the robot arm, allowing multi-directional movement of the fourth segment **124**. The wrist joint **122** can include but is not limited to a ball-and-socket joint, a multi-directional gimbal, a spherical bearing, or any other mechanism supporting multi-directional movement. The wrist joint **122** can be configured as a permanent joint or as a detachable mechanism. In an embodiment, the robot arm may include multiple wrist joints **122**.

(39) A terminal joint **126** is connected to the distal end of the fourth segment **124**. The terminal joint is connected to the proximal end of a translation guide **128**, allowing rotation about the horizontal axis of the terminal joint and moving the translation guide **128** rotationally along its vertical plane. In an embodiment, the terminal joint **126** is configured to rotate about its vertical axis allowing the translation guide **128** to move along its horizontal plane. The terminal joint **126** can be configured as a permanent joint or as a detachable mechanism.

(40) In an embodiment, movement of the robotic arm joints is powered by a collection of actuators. The actuators can include but are not limited to DC or servo motors, pneumatic drive, hydraulic systems, or any other mechanism capable of precisely moving the robotic arm. For example, each actuator can be coupled to one or more gears that are configured to move the arm segments.

(41) In an embodiment, a complex of sensors is coupled to the arm assembly **106** to measure and control arm function and the environment. To measure and maintain the position of the arm assembly **106**, the present invention may include one or more sensors for gauging the orientation of the arm. These sensors may be connected to one or more sections of the arm assembly **106** and/or located at the distal end of the arm assembly **106**. Non-limiting examples of such sensors include

encoders, inertial measurement units (IMUs), force and torque sensors, optical motion tracking systems, ultrasonic sensors, LIDAR sensors, RADAR sensors, proximity sensors (capacitive or inductive), and vision systems with depth-sensing cameras.

(42) In some embodiments, one or more additional sensors may be coupled to the arm assembly **106** at a position to allow the force exerted by and on the robotic arm to be measured. In an embodiment, sensors may be coupled to the arm assembly **106** to track the environment surrounding the device **101**. In an embodiment, a camera can be connected to the distal end of the robot arm assembly **106** to track position during translation of the needle.

(43) Referring now to FIG. **3**, embodiments of the present invention further include a needle translation mechanism **130** mechanically coupled to the distal end of robot arm assembly **106** for communication with and control of the needle **102**. In some embodiments, needle translator **130** includes a channel configured to receive the translation guide **128**, thereby allowing needle translator **130** to move along the length of the translation guide **128**.

(44) In some embodiments, the needle translator **130** includes a sliding/translation mechanism that is configured to cause translation of the needle translator **130** about the length of the translation guide **128**. Non-limiting examples of translation mechanisms include lead screws, ball screws, rack and pinion systems, linear actuators, pneumatic cylinders, hydraulic cylinders, belt and pulley systems, chain drives, cam-driven mechanisms, and magnetic linear drives. However, alternative mechanisms can be used to effect translation of the needle translator **130** about the length of the translation guide **128**.

(45) In some embodiments, the translation mechanism is configured to engage the detents **133** along the arm's length. These detents **133** function as locking channels, allowing the needle translator **130** to be secured at various positions along the translation guide **128** depending on the procedural requirements. A built-in locking mechanism, including but not limited to spring-loaded pin or a clamping latch, ensures that the needle translator **130** remains firmly in place at a chosen detent **133**. However, it is contemplated that alternative mechanisms for securing the arm assembly **106** in place during operation can be used.

(46) In some embodiments, the needle translator **130** contains a needle rotator that is configured to cause rotation of the needle **102** about the longitudinal axis of the needle. In some embodiments, the needle rotator includes a gearing system in operable communication with a motor (all housed within the needle translator **130**). The needle rotator includes an exposed gear **132** that is configured to mesh with the needle **102** or a needle hub to cause rotation of the needle **102**.

(47) In some embodiments, the needle rotator is configured to cause both axial translation and oscillatory rotation of the needle **102**. To effect simultaneous rotational oscillation and axial translation, the needle rotator may include a secondary motor that engages a gear mechanism to cause back and forth translation of the needle as the needle rotates. Non-limiting examples include a gear train or eccentric cam mechanism, enabling controlled high-frequency back-and-forth translation of the needle during rotation to enhance tissue extraction. Alternatively, the needle translator **130** is configured to translate in an axial direction as the needle is rotated by the needle rotator.

(48) In some embodiments, the needle translator **130** further includes an extended support structure **134** that is positioned perpendicularly to the translation guide **128** and provides a point of attachment for the proximal hub **107**. The exposed gear **132** is mechanically coupled to the extended support structure **134** of the needle translator **130** and extends out of the extended support structure **134** to operably engage the needle **102** when the proximal hub **107** is secured to the needle translator **130**.

(49) The needle translator **130** may also include a mounting mechanism **136** to securely receive the proximal hub **107**. As depicted, the mounting mechanism **136** is in the form of a male dovetail guide system that is configured to engage a corresponding dovetail receipt **152** for connecting to the proximal hub **107**. However, alternative mechanisms and shapes may be used to ensure that the

mounting mechanism **136** can securely and/or temporarily engage the proximal hub **107**.

(50) In some embodiments, the extended support structure **134** further includes a recess **140** located at an upper end of the extended support structure **134**. The recess **140** is shaped and sized to receive a portion of the needle **102** and/or the needle hub. In addition, recess **140** provides sufficient clearance to allow for rotation of the needle **102** relative to the extended support structure **134**.

(51) The terminal joint **126** may include a securing mechanism **142** to receive and secure the removable distal hub **108** prior to operation. The securing mechanism **142** may include but is not limited to a key-hole slot or a dovetail guide. However, it is contemplated that alternative mechanisms for securing the removable distal hub **108** can be used. In an embodiment, the securing mechanism **142** is positioned in line with the extended support structure **134** of the needle translator **130** that houses the needle rotator.

(52) In some embodiments, securing mechanism **142** has a design or shape to prevent rotation of the distal hub **108** relative to the terminal joint **126**. The terminal joint is configured to rotate relative to other portions of the arm assembly **106**, but not relative to the distal hub **108**. Thus, the connection of the distal hub **108** to the securing mechanism may be in a form that prevents relative rotation of the distal hub **108** with respect to the terminal joint **126**.

(53) Now referring to FIG. 4, the present invention includes one or more motors **400**, such as any of the motors described herein and known in the art, and one or more controllers **402**, such as any of the controllers described herein and known in the art to operably control the motors. While the figure depicts a single motor and a single controller, it should be understood that the present invention may include multiple motors and/or multiple controllers to control the location of the arm segments, the translation of the needle, and/or the rotation of the needle. For the sake of brevity, the description related to the communication of the controllers and motors will reference a single controller and a single motor.

(54) Motor **400** is in operable communication with a controller **402**. The present invention may also include one or more motor switches **404**. These components are in operable communication with each other and one or more batteries **406**.

(55) In some embodiments, controller **402** is configured to operate the control the motors **400** through user engagement of the user interface and/or through programming instructions. FIG. 5 illustrates an embodiment in which a position sensor **508** is operably engaged with a motor **500** and/or a controller **502**. A motor switch **504** such as any of the motor switches described herein can be operated according to principles discussed above to electrically couple the controller **502** to one or more batteries **506**. A position sensor **508**, such as but not limited to a Hall sensor or any other sensor described herein or known in the art, outputs a signal representative of a position of the motor **500** or any other part of the arm assembly or needle and sends the signal to the controller **502**. Using the position sensor **508**, the controller **502** can operably control the motor **500** to locate or rotate the needle as needed to achieve a desired result.

(56) FIG. 6 illustrates logic embodied in or communicable with the controller **402**, **502**.

Commencing at state **600**, the motor is energized. When a time period is determined to have elapsed at state **602**, the direction of rotation of the rotor of the motor is reversed at state **604**. The logic loops back to state **602** to continue to reverse the direction of rotation at various time periods, which may be predetermined in some embodiments, until the motor is deenergized, to cause the motor to oscillate during energization.

(57) When a position sensor **508** is implemented as shown in FIG. 5, the logic of FIG. 7 may be employed by the controller. The motor is energized at block **700**, and upon determination at state **702** based on the signal from the position sensor that a position of the motor has been attained, the direction of rotation of the motor is reversed at state **704**. Note that the position at which rotation is reversed may be multiples of a particular angle so that the motor is reversed after one or multiple rotations. For example, the rotor position for reversal may be at 0 degrees, 720 degrees (two full rotations), then again at 1440 degrees (after another two full rotations), etc. Alternatively, the rotor

position for reversal may be every 180 degrees, i.e. less than a complete rotor rotation. The logic loops back from state **704** to state **702** to continue to reverse the direction of rotation until the motor is deenergized to cause the motor to oscillate during energization.

(58) Referring now to FIG. **8**, the present invention may employ a needle assembly **802** including an elongated needle **102** with a sharp cutting edge **803** to facilitate cutting tissue when the needle **102** is advanced into tissue and rotated. The needle **102** may be a hollow metal hypodermic needle of a size of between 18 to 27 gauge, and more particularly 22-25 gauge with a cutting edge **803**, and with a length of 1200 mm to 1300 mm. The cutting edge **803** may include bevels, serrations, or other geometries to optimize its ability to cut through biological material during oscillatory or translational motion.

(59) The needle assembly **802** may be entirely disposable, including the needle **102**, a stylet **804**, a luer hub **805**, and an O-ring **806**-packaged in sterile, single-use kits. The needle **102** itself is constructed from medical-grade stainless steel or titanium to ensure strength, durability, and biocompatibility. In an alternate embodiment, the components may be reusable and constructed from autoclavable materials suitable for repeated use.

(60) The needle assembly **802** further includes a needle gear **807** having gear teeth disposed around the needle gear **807**. In some embodiments, the needle gear **807** could be detachable from the luer hub **805**. The gear teeth are configured to meshingly engage the gear wheel **132** that protrudes from the needle translator **130**. As a result, the motor supported in the needle translator **130** rotates the geared wheel **132** which rotates the needle **102**. The needle **102** has the capacity for rotational speeds between sixty (60) revolutions per minute (RPM) to three hundred fifty (350) RPM, inclusive.

(61) The O-ring **806** of the needle assembly **802** surrounds the portion of the luer hub **805** that sits within the interior of the geared wheel **807**. The connection of the geared wheel **807** to the luer hub **805** is a friction or press fit connection facilitated by an O-ring **806**. The luer hub **805** is constructed with a groove to receive the O-ring **806** and maintain its position. This configuration allows the luer hub **805** to remain static while the geared wheel **807** and needle **102** rotate to maintain an airtight seal between the geared wheel **807** and the luer hub **805**.

(62) The hollow interior of the needle **102** establishes a collection chamber. In an alternative embodiment, the collection chamber includes a separate, removable tissue chamber, which can be detached for direct sample retrieval without disassembling the entire system. A continuous fluid passageway is formed by the hollow interior through the removable distal hub, removable proximal hub, and luer hub. In an embodiment, the needle assembly **802** includes a valve, such as but not limited to a slide valve or stopcock, to lock vacuum in the tissue chamber and occlude tissue from entering the suction tubing. The valve can be embedded within the cylinder supporting the luer hub **805** and O-ring **806**.

(63) A stylet **804** is secured to the proximal end of the needle assembly **802**, opposite the cutting edge **803**. The stylet **804** is a thin, solid wire inserted inside the needle **102** to maintain the needle's rigidity and prevent tissue from obstructing the tissue chamber during insertion. Once the device is assembled, the stylet **804** is removed and the suction tubing **109** is attached.

(64) Finally, the needle assembly **802** includes a sheath **808** that surrounds the needle **102**, providing additional stability and ensuring safe handling and operation during medical procedures. The sheath **808** is constructed from biocompatible materials, such as but not limited to medical-grade stainless steel or autoclavable polymers, designed to withstand high-speed oscillatory forces and axial translation while maintaining structural integrity.

(65) Referring now to FIG. **9** proximal hub **107** may be removably coupled to the needle translator **130** to securely hold the needle **102** and integrate with the needle translator **130** and gear **132**. In an embodiment, the proximal hub **107** may be is a semi-rounded cylinder with at least one of an upper portion **144** and a lower portion **146** having associated widths and extending horizontally toward the needle translator **130**. The upper portion **144** and lower portion **146** function to removably

attach the proximal hub **107** to the needle translator **130**. The upper portion **144** includes a central opening **148** configured to receive the needle assembly **802**. The outer casing **150** of the proximal hub is constructed from medical-grade stainless steel or autoclavable high-performance polymers, such as PEEK, to withstand sterilization cycles and ensure a long operational life. In disposable embodiments, the hub may be made from lightweight polycarbonate for single-use applications in sterile environments. Its internal surfaces are designed with smooth contours to prevent material buildup and facilitate cleaning, while external components include minimal sharp edges to reduce the risk of accidental contamination during handling.

(66) The upper portion **144** and lower portion **146** include an attachment mechanism **152**, such as but not limited to dovetail guides, to facilitate secure, removable attachment to the needle translator **130**. The attachment mechanism **152** allows the proximal hub **107** to slide into position along matching channels on the needle translator **130**, locking it firmly in place to prevent any lateral or rotational movement during operation. In another embodiment, the attachment mechanism **152** includes but is not limited to keyed slots or magnetic couplings to securely fit the removable proximal hub **107** to the needle translator **130**. However, it is contemplated that alternative attachment mechanisms can be used.

(67) The upper portion **144** includes an internal luer hub retention latch **154** configured to receive the luer hub **805** of the needle assembly **802**. The retention latch **154** includes a through-hole in line with the central opening **148** of the proximal hub **107**, which allows the needle assembly **802** to fit inside the proximal hub **107**. The through-hole has a diameter sufficient to securely insert the needle gear **807**. The locking mechanism of the retention latch **154** may include but is not limited to a spring-loaded clamp, friction, snap-fit, or any alternative design to easily insert and remove the needle assembly. The retention latch **154** may be constructed from durable materials capable of repeated sterilization such as but not limited to stainless steel, titanium, or high-strength polymers like PEEK.

(68) To maintain a sterile and airtight seal, the latch **154** is configured to engage with the O-ring **806** at the interface with the luer hub **805**. The O-ring **806** ensures no vacuum leaks occur during operation, preserving the integrity of the vacuum system.

(69) An anti-buckling tube **156** attached to the distal end of the proximal hub **107** ensures the stability and proper axial alignment of the needle assembly **802** during operation. The anti-buckling tube **156** is constructed from durable materials with an ability to withstand repeated sterilization cycles. The inner surface of the tube **156** is polished or coated with a low-friction material, such as but not limited to PTFE (Teflon), to minimize resistance as the needle **102** moves axially or oscillates within the tube **156**. This surface treatment also prevents the buildup of biological debris or contaminants during operation, facilitating cleaning and maintenance.

(70) The anti-buckling tube **156** is hollow and cylindrical, with an inner diameter slightly larger than the needle's **102** outer diameter. This tight clearance ensures that the needle **102** is fully supported while allowing for unimpeded axial translation and oscillation. The length of the tube **156** is calibrated to match the distance between the proximal hub **107** and the distal hub **108**. This length prevents buckling across the entire unsupported span of the needle **102**.

(71) Internally, the proximal hub **107** includes a receiver **158** that is configured to receive the needle gear **807**. The receiver **158** secures the needle assembly **802** and allows the needle gear **807** to be in mechanical communication with the gear **132** of the needle translator **130**. The receiver **158** aligns with corresponding needle gear **807** in the needle assembly **802**. The receiver **158** may be constructed from materials such as but not limited to hardened stainless steel or high-strength alloys, chosen for their durability and resistance to wear under repeated high-frequency operation. The internal surfaces of the receiver **158** may be coated with low-friction materials, such as but not limited to PTFE (Teflon) or ceramic composites, to enhance efficiency and minimize heat generation during prolonged use.

(72) FIG. **10** shows a removable distal hub **108** mechanically connected to the terminal joint **126** on

the robot arm assembly **106** that is configured to secure, guide, and stabilize the needle **102** during a medical procedure. In one embodiment, the distal hub **108** includes a securing mechanism **160**, such as but not limited to a keyed mating system. This securing mechanism **160** uses alignment features, such as but not limited to dovetail slots or bayonet-style connectors, to ensure secure and repeatable attachment to the arm assembly **106**. The securing mechanism **160** may include spring-loaded pins or locking tabs to provide quick-release functionality, enabling operators to detach or reattach the hub without requiring specialized tools.

(73) The outer case **161** of the distal hub **108** may be a low-profile, smoothed surface to protect internal components, ease cleaning, and avoid damage or unwanted contact with the environment that could interrupt medical procedures. The outer casing **161** may be constructed from medical-grade materials, such as but not limited to anodized aluminum, stainless steel, or high-performance polymers like PEEK, ensuring durability under repeated use and compatibility with autoclave sterilization cycles. In another embodiment, the distal hub **108** is disposable.

(74) An extended portion **162** of the distal hub **108** extends proximally toward the needle **102** and is attached to a guide tube **164**. The guide tube **164** is configured to have a slightly larger diameter than the anti-buckling tube **156** extending from the lower extended portion **162** of the removable proximal hub **107**. Once assembled, the anti-buckling tube **156** is positioned with the free end sitting inside of the free end of the guide tube **164**.

(75) The guide tube **164** is constructed from high-strength, low-friction materials such as but not limited to polished stainless steel, PTFE (polytetrafluoroethylene), or ceramic-coated alloys, to withstand repeated rounds of sterilization and maintain performance over extended use. In one embodiment, the guide tube **164** and proximally extended portion **162** of the distal hub **108** comprise a singular, contiguous element of the device **101**. In another embodiment, the guide tube **164** is independent from the distal hub **108** and is attached via a secure and vibration resistant connection, such as but not limited to a threaded connection, press-fit collar, or locking clamp. In an embodiment, the guide tube **164** may also include an integrated sealing system, such as but not limited to rings or gaskets, at the junctions where it interfaces with the distal hub **108** and anti-buckling tube **156**.

(76) A sheath clamping latch **166** is internally integrated into the proximally extended portion **162** of the distal hub **108**, positioned perpendicularly to the vertical axis of the extended portion **162**. An opening in the center of the sheath clamping latch **166** is oriented to receive and secure the sheath **808** surrounding the needle **102** and allow the needle **102** to pass through the distal hub **108**. The clamping latch **166** may include but is not limited to a sliding collar, twist-lock mechanism, or spring clamp to hold the sheath firmly in place during operation while allowing for easy removal when replacing the sheath.

(77) The clamping latch **166** is constructed from autoclavable materials, such as but not limited to anodized aluminum, stainless steel, or high-strength polymers like PEEK, to ensure durability and compatibility with repeated sterilization cycles. For disposable embodiments, the latch can be made from medical-grade plastics, such as polycarbonate or polypropylene, offering cost-efficient single-use solutions. In some embodiments, the clamping latch **166** includes an adjustable locking system to accommodate sheaths of varying diameters or lengths.

(78) An endoscope **168** integrated into the removable distal hub **108**, positioned opposite the guide tube **164**, and provides real-time visual feedback during medical procedures. The endoscope **168** is housed within a dedicated channel or sleeve inside the distal hub **108**, running parallel to the needle assembly **802**. The endoscope **168** includes an optical system, such as but not limited to a miniature camera module or fiber optic bundle at the distal end, designed to capture high-resolution images or videos. In an embodiment, the endoscope **168** may include features such as zoom functionality, autofocus, or panoramic views, depending on the procedural requirements.

(79) FIG. **11** shows an exploded view of the needle assembly **802**, proximal hub **107**, and distal hub **108** prior to assembly. To assemble the overall device **101**, the needle assembly **802** is inserted into

the proximal hub **107** so that the needle assembly **802** passes through to the anti-buckling tube **156** of the proximal hub **107**. The anti-buckling tube **156** is inserted into the guide tube **164**, creating a continuous passage for the needle assembly **802** from the proximal hub **107** to the endoscope **168**. The proximal hub **107** and distal hub **108** are then loaded onto the robotic arm assembly **106** prior to operation, as shown in FIG. **12**.

(80) Continuing the description for FIG. **12**, suction tubing **109** connects distally to the needle assembly **802** and proximally to the main body **103**. The tubing **109** is constructed from medical-grade materials such as but not limited to silicone, thermoplastic elastomers (TPE), or other suitable material, chosen for their biocompatibility, flexibility, and resistance to wear and tear. The tubing **109** construction includes an inner layer for maintaining suction flow integrity and an outer layer for mechanical protection.

(81) The tubing **109** is typically manufactured in lengths of 1 to 3 meters, with an inner diameter ranging from 3 to 5 millimeters to balance suction efficiency and flexibility. For applications requiring higher suction rates, larger diameters may be used, while smaller diameters are preferred for more delicate procedures. In another embodiment, reinforced versions of the tubing **109** may include an outer metal wire helix or ribbing to prevent kinking or collapse under vacuum pressure, ensuring consistent suction performance during operation. This reinforcement also enhances resilience when bent or compressed within the robotic arm's constrained pathways.

(82) FIG. **13** shows an x-ray view of the proximal hub **107** attached to the needle translator **130**. In the configuration shown, the needle gear **807** is secured in the receiver **158** of the proximal hub **107**. The luer hub retention latch **154** engages the luer hub **805** to secure the needle assembly **802**. The distal end of the suction tubing **109** is connected to the proximal end of the needle assembly **802** where the stylet **804** was attached prior to overall assembly of the device **101**.

(83) The proximal end of the tubing **109** attaches to the main body **103** through a quick-connect/disconnect coupling system that facilitates rapid assembly and disassembly without requiring specialized tools. In an embodiment, this coupling to the main body **103** may incorporate a one-way valve, which automatically seals when the tubing **109** is detached, preserving system pressure and preventing contamination.

(84) In an embodiment, the suction tubing **109** is configured to be disposable and provided in sterile packaging to ensure contamination-free use in medical procedures. The disposable suction tubing **109** connects to the main body **103** and the needle assembly **802** through reusable hardware designed for secure, sterile, and easy-to-clean operation. Mechanisms connecting the suction tubing **109** to the device **101** are constructed to withstand repeated sterilization cycles, with smooth, non-porous surfaces to facilitate thorough cleaning and prevent the buildup of contaminants.

(85) FIG. **14** shows a complete x-ray view of the distal portion of the robot arm assembly **106** connected to the proximal hub **107** and distal hub **108**. The needle assembly **802** is secured by the luer hub retention latch **154** in the proximal hub **107**, which in turn is connected to the needle translator **130**. The distal end of the suction tubing **109** is connected to the proximal end of the needle assembly **802**, as previously described. The anti-buckling tube **156** is positioned with its distal end sitting within the diameter of the proximal end of the guide tube **164**. The sheath clamping latch **154** is engaged to secure the sheath **808** and allow the needle assembly **802** to pass through the distal hub **108** to the endoscope **168**.

(86) The device **101** can be configured to employ various microcontrollers (MCUs) and processors. In one embodiment, low to mid-range MCUs like the Arduino family (e.g., ATmega328, ATmega2560) are suitable due to their low power consumption and straightforward programming. More sophisticated configurations may require advanced MCUs, such as the STM32, that offer advanced peripherals like timers, ADCs, and communication interfaces (CAN, PC, SPI). These are ideal for subsystems that manage real-time data from multiple sensors or control robotic arms. Higher-performance MCUs, such as the Texas Instruments Tiva C series or ESP32, can handle more complex operations like multi-axis motor synchronization and wireless communication

through integrated Wi-Fi and Bluetooth. For tasks such as real-time video processing or advanced user interfaces, single-board computers (SBCs) like the Raspberry Pi (4B, CM4) or NVIDIA Jetson Nano/Xavier are highly effective. These processors are capable of handling machine learning and computer vision, making them suitable for systems requiring AI-driven decision-making. Industrial-grade processors, including Intel Core i3/i5 or AMD Ryzen Embedded, provide powerful multi-threading capabilities for teleoperation and real-time data streaming. Additionally, field-programmable gate arrays (FPGAs), such as the Xilinx Zynq or Intel Cyclone series, offer custom hardware acceleration for tasks requiring ultra-fast response times, such as fine needle positioning or high-frequency oscillation control.

(87) The software architecture is built on a combination of microcontroller and single-board computer (SBC) environments. For microcontrollers, development can be carried out using Arduino IDE or PlatformIO, allowing for efficient coding of low-level operations such as motor control and sensor data acquisition. SBCs, such as Raspberry Pi or NVIDIA Jetson, handle higher-level tasks and are programmed in Python, C++, or other comparable languages. These environments can integrate specialized libraries like OpenCV for computer vision, enabling advanced features such as image processing and object tracking. The system's modular software design ensures scalability, allowing new functionalities to be added as needed.

(88) The techniques introduced here can be embodied as special-purpose hardware (e.g. circuitry), as programmable circuitry appropriately programmed with software and/or firmware, or as a combination of special-purpose and programmable circuitry. Hence, embodiments may include a machine-readable medium having stored thereon instructions which may be used to program a computer (or other electronic devices) to perform a process. The machine-readable medium may include, but is not limited to, floppy diskettes, optical disks, compact disc read-only memories (CD-ROMs), magneto-optical disks, ROMs, random access memories (RAMs), erasable programmable read-only memories (EPROMs), electrically erasable programmable read-only memories (EEPROMs), magnetic or optical cards, flash memory, or other type of media/machine-readable medium suitable for storing electronic instructions.

(89) A computer readable signal medium may include a propagated data signal with computer readable program PIN embodied therein, for example, in baseband or as part of a carrier wave. Such a propagated signal may take any of a variety of forms, including, but not limited to, electromagnetic, optical, or any suitable combination thereof. A computer readable signal medium may be any computer readable medium that is not a computer readable storage medium and that can communicate, propagate, or transport a program for use by or in connection with an instruction execution system, apparatus, or device.

(90) Program PIN embodied on a computer readable medium may be transmitted using any appropriate medium, including but not limited to wireless, wire-line, optical fiber cable, radio frequency, etc., or any suitable combination of the foregoing. Computer program PIN for carrying out operations for aspects of the present invention may be written in any combination of one or more programming languages, including an object oriented programming language such as Java, C#, C++, Python, MATLAB, and/or the like and conventional procedural programming languages, such as the "C" programming language or similar programming languages.

(91) Aspects of the present invention are described below with reference to flowchart illustrations and/or block diagrams of methods, apparatus (systems) and computer program products according to embodiments of the invention. It will be understood that each block of the flowchart illustrations and/or block diagrams, and combinations of blocks in the flowchart illustrations and/or block diagrams, can be implemented by computer program instructions. These computer program instructions may be provided to a processor of a general purpose computer, special purpose computer, or other programmable data processing apparatus to produce a machine, such that the instructions, which execute via the processor of the computer or other programmable data processing apparatus, create means for implementing the functions/acts specified in the flowchart

and/or block diagram block or blocks.

(92) These computer program instructions may also be stored in a computer readable medium that can direct a computer, other programmable data processing apparatus, or other devices to function in a particular manner, such that the instructions stored in the computer readable medium produce an article of manufacture including instructions which implement the function/act specified in the flowchart and/or block diagram block or blocks.

(93) The computer program instructions may also be loaded onto a computing device, other programmable data processing apparatus, or other devices to cause a series of operational steps to be performed on the computer, other programmable apparatus or other devices to produce a computer implemented process such that the instructions which execute on the computer or other programmable apparatus provide processes for implementing the functions/acts specified in the flowchart and/or block diagram block or blocks.

Claims

1. A robotic medical device, comprising: a needle assembly, the needle assembly including: a hollow needle; a needle gear at a proximal end of the hollow needle; a luer hub with an O-ring, wherein the needle is configured to rotate relative to the luer hub; a robotic arm, the robotic arm including: multiple joints providing at least three degrees of freedom; a translation guide and a needle translator configured to axially translate the needle assembly about a length of the translation guide; a motor configured to operably engage the needle gear and thereby rotate the needle about a longitudinal axis of the needle; a tissue chamber being established at least in part by a hollow interior of the needle and in fluidic communication with a vacuum source via a suction tube; a proximal hub configured to house the proximal end of the needle and temporarily attach to the needle translator; a distal hub configured to be temporarily coupled to the robotic arm at a distal location relative to the proximal hub; wherein the needle translator is configured to axially translate the proximal hub and needle relative to the distal hub and rotate the needle about the longitudinal axis of the needle and the tissue chamber is subject to the vacuum.

2. The device of claim 1, wherein the proximal hub comprises: a removable attachment mechanism; a luer hub retention latch; an interface for transferring rotational motion from the motor to the needle gear; and an anti-buckling tube extending towards the distal hub, wherein the needle passes through the anti-buckling tube.

3. The device of claim 1, wherein the distal hub comprises: a removable attachment mechanism; a sheath clamping latch; a guide tube; and an endoscope.

4. The device of claim 1, wherein the needle assembly and suction tube are disposable.

5. The device of claim 1, wherein the robotic arm includes a detachable joint mechanism with alignment pins and a quick-release locking collar to facilitate removal and reattachment.

6. The device of claim 1, further comprising sensors, including: a position sensor configured to monitor movement of the robotic arm; a pressure sensor configured to monitor vacuum levels within the tissue chamber; and a force-torque feedback sensor configured to monitor the rotation and translation of the needle.

7. A robotic medical device, comprising: a needle assembly, the needle assembly including: a hollow needle; a needle gear disposed proximate to a proximal end of the hollow needle; a luer hub with an O-ring, wherein the needle is configured to rotate relative to the luer hub; a robotic arm, the robotic arm including: a translation guide and a needle translator configured to axially translate the needle assembly about a length of the translation guide; a proximal hub configured to house the needle and temporarily attach to the needle translator; a motor configured to operably engage the needle gear and thereby rotate the needle about a longitudinal axis of the needle; a distal hub configured to be temporarily coupled to a terminal joint of the robotic arm, wherein the distal hub includes a passage for receiving the needle; a tissue chamber being established at least in part by a

hollow interior of the needle and in fluidic communication with a vacuum source via a suction tube; wherein the needle translator is configured to axially translate the proximal hub and needle relative to the distal hub while the needle is rotated relative to the longitudinal axis of the needle and the tissue chamber is subject to a vacuum.

8. The device of claim 7, wherein the proximal hub comprises: a removable attachment mechanism; a luer hub retention latch; and an interface for transferring rotational motion from the motor to the needle gear.

9. The device of claim 7, further including an anti-buckling tube extending at least partially between the distal hub and the proximal hub through which the needle passes, wherein the anti-buckling tube has greater rigidity than the needle.

10. The device of claim 7, wherein the distal hub comprises: a removable attachment mechanism; a sheath clamping latch; a guide tube; and an endoscope.

11. The device of claim 7, wherein the needle assembly and suction tube are disposable.

12. The device of claim 7, wherein the robotic arm includes a detachable joint mechanism with alignment pins and a quick-release locking collar to facilitate removal and reattachment.

13. The device of claim 7, further comprising sensors, including: a position sensor configured to monitor movement of the robotic arm; a pressure sensor configured to monitor vacuum levels within the tissue chamber; and a force-torque feedback sensor configured to monitor the rotation and translation of the needle.

14. A robotic medical device, comprising: a needle assembly, the needle assembly including: a hollow needle; a needle gear disposed proximate to a proximal end of the hollow needle; a robotic arm, the robotic arm including: a translation guide and a needle translator; a proximal hub configured to house the proximal end of the needle and temporarily attach to the needle translator; a motor configured to operably engage the needle gear and thereby rotate the needle about a longitudinal axis of the needle; a distal hub configured to be temporarily coupled to a terminal joint of the robotic arm, wherein the distal hub includes a passage for receiving the needle; wherein the needle translator is configured to translate the needle through the distal hub and translate the proximal hub towards the distal hub; a tissue chamber being established at least in part by a hollow interior of the needle and in fluidic communication with a vacuum source; wherein the needle can be axially translated and rotated relative to the longitudinal axis of the needle while the tissue chamber is subject to a vacuum.

15. The device of claim 14, wherein the proximal hub comprises: a removable attachment mechanism; a luer hub retention latch; and an interface for transferring oscillatory and translational motion from the motor to the needle gear.

16. The device of claim 14, further including an anti-buckling tube extending at least partially between the distal hub and the proximal hub, wherein the needle passes through the anti-buckling tube and the anti-buckling tube has greater rigidity than the needle.

17. The device of claim 14, wherein the distal hub comprises: a removable attachment mechanism; a sheath clamping latch; a guide tube; and an endoscope.

18. The device of claim 14, further comprising sensors, including: a position sensor configured to monitor movement of the robotic arm; a pressure sensor configured to monitor vacuum levels within the tissue chamber; and a force-torque feedback sensor configured to monitor the rotation and translation of the needle.
